Astha Yadav

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Objective

Physics Graduate aspiring to research in Quantum Communication - intertwining Quantum Physics with cryptography and network security.

Technical Skills

- **Quantum Simulations:** Qiskit, NetSquid, QKD Vulnerability Modelling, Decoy-State Protocols
- **Programming & Tools:** Python, MATLAB, SQL, HTML/CSS, Git/GitHub, LaTeX
- **Data Science & Numerical Methods:** NumPy, Pandas, Scikit-learn, Matplotlib, Graph Optimization, Euler/RK4 Integration, Finite Difference Methods
- **Cybersecurity & Network Infrastructure:** Network Security Infrastructure, Cyber Forensics, Statistical Anomaly Detection

Work Experience

- **Intern - Quantum Networks** *August 2026 - Present*
  Manager: Haris Ansari Quantum AI Global, India
  - Working on the backend codes of upcoming Quantum Network simulator by the company.
  - Reviewing and working on the quantum networks Learning Material accompanying the simulator.
  - Thorough testing of the simulator focusing on physics bugs.

- **Quantum Networks Summer Intern at SAG, DRDO** *June 2026 - July 2026*
  Project guide: Devendra KY, Scientist (F) SAG, DRDO
  - Modified and extended the CHSH algorithm from my bachelor's thesis to 7, 15 and 50 node meshes. Analysing shortcomings of Simple Routing Algorthitms.
  - Studying key papers on routing protocols, purification metrics and path selection policies.
  - Working on purification based routing algorithm implementation on a dynamically weighted 15-node mesh.

- **Co-Author & Research Assistant in Quantum Chemistry Simulation** *March 2026 - May 2026*
  Collaborative Research Project Fujitsu Quantum Simulator Challenge 2026
  - Worked in a multidisciplinary team to execute 16-qubit simulations on IQM's Sirius 24-qubit superconducting processor, mapping the 2D potential energy surface (32×32 grid) of H_2O
  - Co-authored a research article demonstrating the integration of Density Matrix Embedding Theory (DMET) in fragment-based molecular simulations to scale them according to industrial limits while maintaining chemical accuracy
  - Formulated a comprehensive literature review on Sample-based Quantum Diagonalization (SQD) frameworks, establishing the theoretical baseline for the molecular electronic structure simulations
  - We evaluated the ansatz expressivity and gate-depth trade-offs between hardware-efficient LUCJ and chemically motivated LCNot-UCCSD.

Academic Research Projects

- **Bachelor's Thesis in QKD and QNetworks** *August 2025 - May 2026*
  Advisor: Prof. Bipin Singh Koranga Kirori Mal College
  - Thesis has two investigative projects: 1) simulation and modelling of errors, vulnerabilities and mitigation strategies in BB84, and 2) modelling E91-based entanglement network and a simple quantum routing algorithm.
  - Developed BB84 protocol simulations in Python to establish error-free baseline QBER, quantify channel attenuation and single-photon detector (SPD) noise, and model pulse vulnerabilities under Photon Number Splitting (PNS) attacks.

- Evaluated error mitigation strategies for BB84 using decoy-state protocols, and developed a Multi-Layer Perceptron (MLP) machine learning model in Python to eliminate environmental drift in CVQKD systems.
- Learned quantum networking concepts, including entanglement swapping topologies, quantum memory, state purification mechanics, and localized fidelity tracking.
- Built discrete-event simulations in NetSquid to model a) quantum teleportation, b) E91-CHSH verification loops, and c) Centralized CHSH Bottleneck Routing algorithm utilizing max-min graph optimization for a simple 5-node mesh.
- Skills: Quantum Key Distribution, Quantum Networks, Python, Qiskit, NetSquid, Machine Learning

Competitive Achievements

- **Winner of BB84 Challenge** | IBM Qiskit Fall Fest '25 (PES University) *Nov 2025*
- **Participant** | MIT iQuHack 2026 (IonQ Entanglement Distillation Challenge) *Feb 2026*

Technical Certifications & Diplomas

- **QWorld Diplomas:** **QSilver** (March 2026) | **QBronze** (July 2024)
- **FDP MeitY:** Quantum Computing & Post-Quantum Cryptography *Feb 2025*
- **IBM Quantum:** Qiskit Global Summer School 2025 Badge *Aug 2025*
- **WISER Summer Program:** Quantum Algorithms | Non-Linear Problems *July 2025*
- **Short Courses:** IQM Quantum School | TryHackMe Cybersec Challenge *Dec 2025*

Education

- **B.Sc. (Hons. with Research) in Physics, with Minor in Computer Science** *November 2022 - June 2026*
  Kirori Mal College, University of Delhi Delhi, India
  - Percentage: 72%
- **Senior-Seconday/12th Grade** *April 2022*
  CBSE Board Delhi, India
  - Percentage: 77.6%
- **Secondary/10th Grade** *April 2020*
  CBSE Board Delhi, India
  - Percentage: 96%

Additional Achievements

Languages: English (College Medium of Instruction), French (A1 Certificate), Hindi & Awadhi (Native)
Professional Memberships: American Physical Society, QIndia, Girls in Quantum, Quantum Flagship
Leadership Roles: Lead Communications Systems Semester Project (Sem-5), Student Council of Film Society, Directed College Media Projects

Physics Hons Coursework

Theoretical Physics: Classical Mechanics (Lagrangian & Hamiltonian), Quantum Mechanics I & II (Hilbert Spaces, Angular Momentum, Density Matrices), Advanced Quantum Mechanics (Time-Dependent Perturbation, Fermi's Golden Rule, Scattering Theory, Born Approximation, Variational Methods), Statistical Mechanics & Advanced Statistical Mechanics (Grand Canonical Ensembles, Ising Model, Quantum Density Matrices), Electromagnetic Theory (Maxwell's Eqns, Wave Guides), Special Relativity, and Modern Physics.
Mathematical & Computational: Mathematical Physics I-III (Complex Analysis, PDEs, Fourier Transforms), Linear Algebra, Numerical Methods, Bayesian Statistics, Python libraries (Pandas, Seaborn, NumPy, Scikit-learn, Matplotlib)
Applied & Experimental: Solid State Physics (Brillouin Zones, Hall Effect), Atomic & Molecular Physics, Analog & Digital Electronics (BJTs, Op-Amps, Logic Circuits), Communication Systems, Waves and Optics, and Thermal Physics.
Ancillary: DBMS, Operating Systems, Web Development, and Research Methodology (Ethics & IPR).

Computer Science Coursework

Core Systems & Architecture: Operating Systems (Process Scheduling, Paging, Shell Scripting), Database Management Systems (ER Modeling, SQL, 1NF/2NF/3NF Normalization), and Web Programming (HTML/CSS, JavaScript, jQuery, JSON).
Data Structures & Algorithms: Data Structures Using Python (Linked Lists, Stacks, Queues, Binary Search Trees, Binary Heaps), Growth of Functions, and Recurrence Relations.
Intelligence & Language Modeling: Machine Learning (PCA, Regression, SVM, Back-propagation Neural Networks, Clustering) and Natural Language Processing (TF-IDF, Word2vec, RNNs/LSTMs, Transformers, BERT, GPT).
Security & Forensics: Information Security (Symmetric/Public-Key Cryptography, PKI, RBAC/ABAC, SQL Injection, Firewalls/DDoS Defenses, Cloud/IoT Security) and Cyber Forensics (Evidence Recovery, Hive File Analysis, Autopsy/EnCase/FTK, IT Laws).